

FOURIER SERIES

Periodic phenomena involving waves, rotating machines (harmonic motion), or other repetitive driving forces are described by periodic functions. Fourier series are a basic tool for solving ordinary differential equations (ODEs) and partial differential equations (PDEs) with periodic boundary conditions. Fourier integrals for nonperiodic phenomena are developed in Chapter 20. The common name for the field is **Fourier analysis**.

19.1 GENERAL PROPERTIES

A Fourier series is defined as an expansion of a function or representation of a function in a series of sines and cosines, such as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx. \quad (19.1)$$

The coefficients a_0 , a_n , and b_n are related to $f(x)$ by definite integrals:

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(s) \cos ns \, ds, \quad n = 0, 1, 2, \dots, \quad (19.2)$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(s) \sin ns \, ds, \quad n = 1, 2, \dots, \quad (19.3)$$

which are subject to the requirement that the integrals exist. Note that a_0 is singled out for special treatment by the inclusion of the factor $\frac{1}{2}$. This is done so that Eq. (19.2) will apply to all a_n , $n = 0$ as well as $n > 0$.

The conditions imposed on $f(x)$ to make Eq. (19.1) valid are that $f(x)$ have only a finite number of finite discontinuities and only a finite number of extreme values (maxima and minima) in the interval $[0, 2\pi]$.¹ Functions satisfying these conditions may be

¹These conditions are **sufficient** but not **necessary**.

called **piecewise regular**. The conditions themselves are known as the **Dirichlet conditions**. Although there are some functions that do not obey these conditions, they can be considered pathological for purposes of Fourier expansions. In the vast majority of physical problems involving a Fourier series, the Dirichlet conditions will be satisfied.

Expressing $\cos nx$ and $\sin nx$ in exponential form, we may rewrite Eq. (19.1) as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \quad (19.4)$$

in which

$$c_n = \frac{1}{2} (a_n - ib_n), \quad c_{-n} = \frac{1}{2} (a_n + ib_n), \quad n > 0, \quad (19.5)$$

and

$$c_0 = \frac{1}{2} a_0. \quad (19.6)$$

Sturm-Liouville Theory

The ODE

$$-y''(x) = \lambda y(x)$$

on the interval $[0, 2\pi]$ with boundary conditions $y(0) = y(2\pi)$, $y'(0) = y'(2\pi)$ is a Sturm-Liouville problem, and these boundary conditions make it Hermitian. Therefore its eigenfunctions, either $\cos nx$ ($n = 0, 1, \dots$) and $\sin nx$ ($n = 1, 2, \dots$), or $\exp(inx)$ ($n = \dots, -1, 0, 1, \dots$), form a complete set, with eigenfunctions of different eigenvalues orthogonal. Since the eigenfunctions have respective values n^2 , those of different $|n|$ will automatically be orthogonal, while those of the same $|n|$ can be orthogonalized if necessary. Defining the scalar product for this problem as

$$\langle f | g \rangle = \int_0^{2\pi} f^*(x) g(x) dx,$$

it is easy to check that $\langle e^{inx} | e^{-inx} \rangle = 0$ for $n \neq 0$, and if we write $\cos nx$ and $\sin nx$ as complex exponentials, it is also easy to see that $\langle \sin nx | \cos nx \rangle = 0$. To make the eigenfunctions normalized, a simple approach is to note that the average value of $\sin^2 nx$ or $\cos^2 nx$ over an integer number of oscillations is $1/2$ (again for $n \neq 0$), so

$$\int_0^{2\pi} \sin^2 nx dx = \int_0^{2\pi} \cos^2 nx dx = \pi \quad (n \neq 0),$$

and $\langle e^{inx} | e^{inx} \rangle = 2\pi$.

The relationships identified above indicate that the eigenfunctions $\varphi_n = e^{inx}/\sqrt{2\pi}$, ($n = \dots, -1, 0, 1, \dots$) form an orthonormal set, as do

$$\varphi_0 = \frac{1}{\sqrt{2\pi}}, \quad \varphi_n = \frac{\cos nx}{\sqrt{\pi}}, \quad \varphi_{-n} = \frac{\sin nx}{\sqrt{\pi}}, \quad (n = 1, 2, \dots),$$

so expansions in these functions have the forms given in Eqs. (19.1) to (19.3) or Eqs. (19.4) to (19.6). Since we know that the eigenfunctions of a Sturm-Liouville operator form a complete set, we know that our Fourier-series expansions of L^2 functions will at least converge in the mean.

Discontinuous Functions

There are significant differences between the behavior of Fourier- and power-series expansions. A power series is essentially an expansion about a point, using only information from that point about the function to be expanded (including, of course, the values of its derivatives). We already know that such expansions only converge within a radius of convergence defined by the position of the nearest singularity. However, a Fourier series (or any expansion in orthogonal functions) uses information from the entire expansion interval, and therefore can describe functions that have “nonpathological” singularities within that interval. However, we also know that the representation of a function by an orthogonal expansion is only guaranteed to converge **in the mean**. This feature comes into play for the expansion of functions with discontinuities, where there is no unique value to which the expansion must converge. However, for Fourier series, it can be shown that if a function $f(x)$ satisfying the Dirichlet conditions is discontinuous at a point x_0 , its Fourier series evaluated at that point will be the arithmetic average of the limits of the left and right approaches:

$$f_{\text{Fourier series}}(x_0) = \lim_{\varepsilon \rightarrow 0} \left[\frac{f(x_0 + \varepsilon) + f(x_0 - \varepsilon)}{2} \right]. \quad (19.7)$$

For proof of Eq. (19.7), see Jeffreys and Jeffreys or Carslaw (Additional Readings). It can also be shown that if the function to be expanded is continuous but has a finite discontinuity in its first derivative, its Fourier series will then exhibit uniform convergence (see Churchill, Additional Readings). These features make Fourier expansions useful for functions with a variety of types of discontinuities.

Example 19.1.1 SAWTOOTH WAVE

An idea of the convergence of a Fourier series and the error in using only a finite number of terms in the series may be obtained by considering the expansion of

$$f(x) = \begin{cases} x, & 0 \leq x < \pi, \\ x - 2\pi, & \pi < x \leq 2\pi. \end{cases} \quad (19.8)$$

This is a sawtooth wave form, as shown in Fig. 19.1. Using Eqs. (19.2) and (19.3), we find the expansion to be

$$f(x) = 2 \left[\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots + (-1)^{n+1} \frac{\sin nx}{n} + \cdots \right]. \quad (19.9)$$

Figure 19.2 shows $f(x)$ for $0 \leq x < 2\pi$ for the sum of 4, 6, and 10 terms of the series. Three features deserve comment.

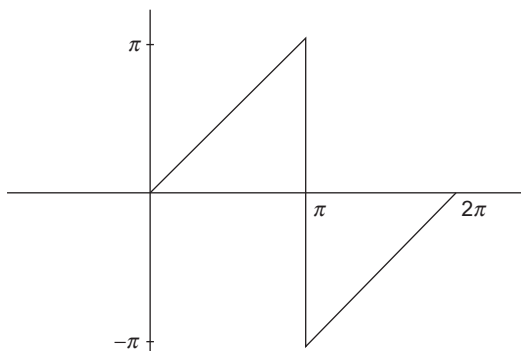
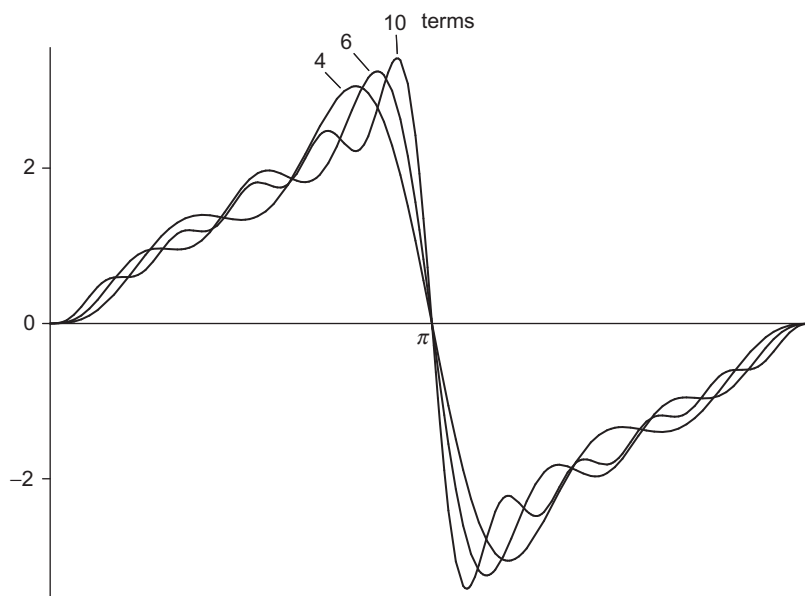


FIGURE 19.1 Sawtooth wave form.

FIGURE 19.2 Expansion of sawtooth wave form, range $[0, 2\pi]$.

1. There is a steady increase in the accuracy of the representation as the number of terms included is increased.
2. At $x = \pi$, where $f(x)$ changes discontinuously from $+\pi$ to $-\pi$, all the curves pass through the average of these two values, namely $f(\pi) = 0$.
3. In the vicinity of the discontinuity at $x = \pi$, there is an overshoot that persists and shows no sign of diminishing.

As a matter of incidental interest, setting $x = \pi/2$ in Eq. (19.9) leads to

$$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} = 2 \left[1 - 0 - \frac{1}{3} - 0 + \frac{1}{5} - 0 - \frac{1}{7} + \cdots \right],$$

thereby yielding an alternate derivation of Leibniz's formula for $\pi/4$, which was obtained by another method in [Exercise 1.3.2](#). ■

Periodic Functions

Fourier series are used extensively to represent periodic functions, especially wave forms for signal processing. The form of the series is inherently periodic; the expansions in [Eqs. \(19.1\) and \(19.4\)](#) are periodic with period 2π , with $\sin nx$, $\cos nx$, and $\exp(inx)$, each completing n cycles of oscillation in that interval. Thus, while the coefficients in a Fourier expansion are determined from an interval of length 2π , the expansion itself (if the function involved is actually periodic) applies for an indefinite range of x . The periodicity also means that the interval used for determining the coefficients need not be $[0, 2\pi]$ but may be any other interval of that length. Often one encounters situations in which the formulas in [Eqs. \(19.2\) and \(19.3\)](#) are changed so that their integrations run between $-\pi$ and π . In fact, it would have been natural to have restated [Example 19.1.1](#) as dealing with $f(x) = x$, for $-\pi < x < \pi$. This of course does not remove the discontinuity or change the form of the Fourier series. The discontinuity has simply been moved to the ends of the interval in x .

In actual situations, the natural interval for a Fourier expansion will be the wavelength of our wave form, so it may make sense to redefine our Fourier series so that [Eq. \(19.1\)](#) becomes

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \quad (19.10)$$

with

$$a_n = \frac{1}{L} \int_{-L}^L f(s) \cos \frac{n\pi s}{L} ds, \quad n = 0, 1, 2, \dots, \quad (19.11)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(s) \sin \frac{n\pi s}{L} ds, \quad n = 1, 2, \dots \quad (19.12)$$

In many problems the x dependence of a Fourier expansion describes the spatial dependence of a wave distribution that is moving (say, toward $+x$) with **phase velocity** v . This means that in place of x we need to write $x - vt$, and this substitution carries the implicit assumption that the wave form retains the same shape as it moves forward.² The individual terms of the Fourier expansion can now be given an interesting interpretation. Taking as an example the term

$$\cos \left[\frac{n\pi}{L} (x - vt) \right],$$

²For waves in physical media, this assumption is by no means always true, as it depends on the time-dependent response properties of the medium.

we note that it describes a contribution of wavelength $2L/n$ (when x increases this much at constant t , the argument of the cosine function increases by 2π). We also note that the period of the oscillation (the change in t at constant x for one cycle of the cosine function) is $T = 2L/nv$, corresponding to the oscillation frequency $\nu = nv/2L$. If we call the frequency for $n = 1$ the **fundamental frequency** and denote it $\nu_0 = v/2L$, we identify the terms for each $n > 1$ in the Fourier series as describing overtones, or **harmonics** of the fundamental frequency, with individual frequencies $n\nu_0$.

A typical problem for which Fourier analysis is suitable is one in which a particle undergoing oscillatory motion is subject to a periodic driving force. If the problem is described by a linear ODE, we may make a Fourier expansion of the driving force and solve for each harmonic individually. This makes the Fourier expansion a practical tool as well as a nice analytical device. We stress, however, that its utility depends crucially on the linearity of our problem; in nonlinear problems an overall solution is not a superposition of component solutions.

As suggested earlier, we have proceeded on the assumption that v , the phase velocity, is the same for all terms of the Fourier series. We now see that this assumption corresponds to the notion that the medium supporting the wave motion can respond equally well to forces at all frequencies. If, for example, the medium consists of particles too massive to respond quickly at high frequency, those components of the wave form will become attenuated and damped out of a propagating wave. Conversely, if the system contains components that resonate at certain frequencies, the response at those frequencies will be enhanced. Fourier expansions give physicists (and engineers) a powerful tool for analyzing wave forms and for designing media (e.g., circuits) that yield desired behaviors.

One question that is sometimes raised is: “Were the harmonics there all along, or were they created by our Fourier analysis?” One answer compares the functional resolution into harmonics with the resolution of a vector into rectangular components. The components may have been present, in the sense that they may be isolated and observed, but the resolution is certainly not unique. Hence many authors prefer to say that the harmonics were created by our choice of expansion. Other expansions in other sets of orthogonal functions would produce a different decomposition. For further discussion, we refer to a series of notes and letters in the *American Journal of Physics*.³

What if a function is not periodic? We can still obtain its Fourier expansion, but (a) the results will of course depend on how the expansion interval is chosen (both as to position and length), and (b) because no information outside the expansion interval was used in obtaining the expansion, we can have no realistic expectation that the expansion will produce there a reasonable approximation to our function.

Symmetry

Suppose we have a function $f(x)$ that is either an even or an odd function of x . If it is even, then its Fourier expansion cannot contain any odd terms (since all terms are linearly independent, no odd term can be removed by retaining others). Our expansion, developed

³B. L. Robinson, Concerning frequencies resulting from distortion. *Am. J. Phys.* **21**: 391 (1953); F. W. Van Name, Jr., Concerning frequencies resulting from distortion. *Am. J. Phys.* **22**: 94 (1954).

for the interval $[-\pi, \pi]$, then must take the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx, \quad f(x) \text{ even.} \quad (19.13)$$

On the other hand, if $f(x)$ is odd, we must have

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx, \quad f(x) \text{ odd.} \quad (19.14)$$

In both cases, when determining the coefficients we only need consider the interval $[0, \pi]$, referring to Eqs. (19.2) and (19.3), as the adjoining interval of length π will make a contribution identical to that considered. The series in Eqs. (19.13) and (19.14) are sometimes called **Fourier cosine** and **Fourier sine** series.

If we have a function defined on the interval $[0, \pi]$, we can represent it either as a Fourier sine series or as a Fourier cosine series (or, if it has no interfering singularities, as a power series), with similar results on the interval of definition. However, the results outside that interval may differ markedly because these expansions carry different assumptions as to symmetry and periodicity.

Example 19.1.2 DIFFERENT EXPANSIONS OF $f(x) = x$

We consider three possible ways to expand $f(x) = x$ based on its values on the range $[0, \pi]$:

- Its power-series expansion will (obviously) have the power-series expansion $f(x) = x$.
- Comparing with Example 19.1.1, its Fourier sine series will have the form given in Eq. (19.9).
- Its Fourier cosine series will have coefficients determined from

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx = \begin{cases} \pi, & n = 0, \\ -\frac{4}{n^2\pi}, & n = 1, 3, 5, \dots, \\ 0, & n = 2, 4, 6, \dots, \end{cases}$$

corresponding to the expansion

$$f(x) = \frac{\pi}{2} - \sum_{n=0}^{\infty} \frac{4}{\pi} \frac{\cos(2n+1)x}{(2n+1)^2}.$$

All three of these expansions represent $f(x)$ well in the range of definition, $[0, \pi]$, but their behavior becomes strikingly different outside that range. We compare the three expansions for a range larger than $[0, \pi]$ in Fig. 19.3. ■

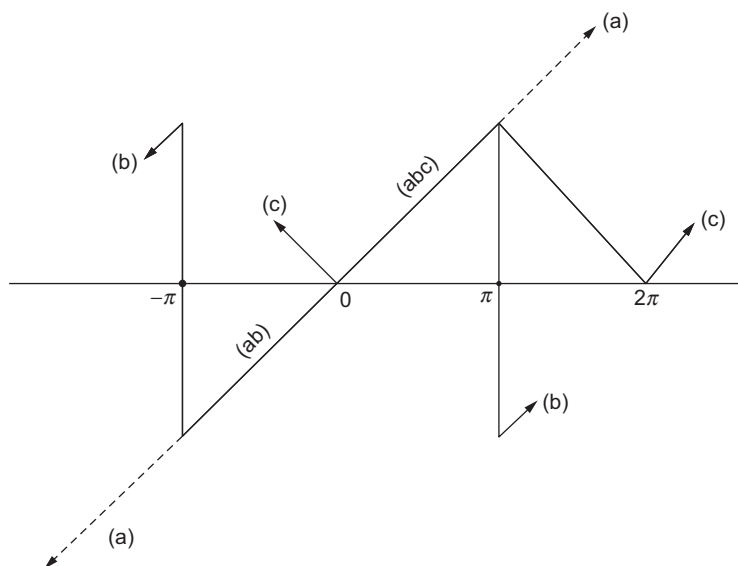


FIGURE 19.3 Expansions of $f(x) = x$ on $[0, \pi]$: (a) power series, (b) Fourier sine series, (c) Fourier cosine series.

Operations on Fourier Series

Term-by-term integration of the series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad (19.15)$$

yields

$$\int_{x_0}^x f(x) dx = \frac{a_0 x}{2} \Big|_{x_0}^x + \sum_{n=1}^{\infty} \frac{a_n}{n} \sin nx \Big|_{x_0}^x - \sum_{n=1}^{\infty} \frac{b_n}{n} \cos nx \Big|_{x_0}^x. \quad (19.16)$$

Clearly, the effect of integration is to place an additional power of n in the denominator of each coefficient. This results in more rapid convergence than before. Consequently, a convergent Fourier series may always be integrated term by term, the resulting series converging uniformly to the integral of the original function. Indeed, term-by-term integration may be valid even if the original series, Eq. (19.15), is not itself convergent. The function $f(x)$ need only be integrable. A discussion will be found in Jeffreys and Jeffreys (Additional Readings).

Strictly speaking, Eq. (19.16) may not be a Fourier series; that is, if $a_0 \neq 0$, there will be a term $\frac{1}{2}a_0x$. However,

$$\int_{x_0}^x f(x) dx - \frac{1}{2}a_0x \quad (19.17)$$

will still be a Fourier series.

The situation regarding differentiation is quite different from that of integration. Here the word is caution. Consider the series for

$$f(x) = x, \quad -\pi < x < \pi. \quad (19.18)$$

We readily found (in [Example 19.1.1](#)) that the Fourier series is

$$x = 2 \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin nx}{n}, \quad -\pi < x < \pi. \quad (19.19)$$

Differentiating term by term, we obtain

$$1 = 2 \sum_{n=1}^{\infty} (-1)^{n+1} \cos nx, \quad (19.20)$$

which is not convergent. **Warning:** Check your derivative for convergence.

For the triangular wave shown in [Fig. 19.4](#) (and treated in [Exercise 19.2.9](#)), the Fourier expansion is

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1, \text{odd}}^{\infty} \frac{\cos nx}{n^2}, \quad (19.21)$$

which converges more rapidly than the expansion of [Eq. \(19.19\)](#); in fact, it exhibits uniform convergence. Differentiating term by term we get

$$f'(x) = \frac{4}{\pi} \sum_{n=1, \text{odd}}^{\infty} \frac{\sin nx}{n}, \quad (19.22)$$

which is the Fourier expansion of a square wave,

$$f'(x) = \begin{cases} 1, & 0 < x < \pi, \\ -1, & -\pi < x < 0. \end{cases} \quad (19.23)$$

Inspection of [Fig. 19.3](#) verifies that this is indeed the derivative of our triangular wave.

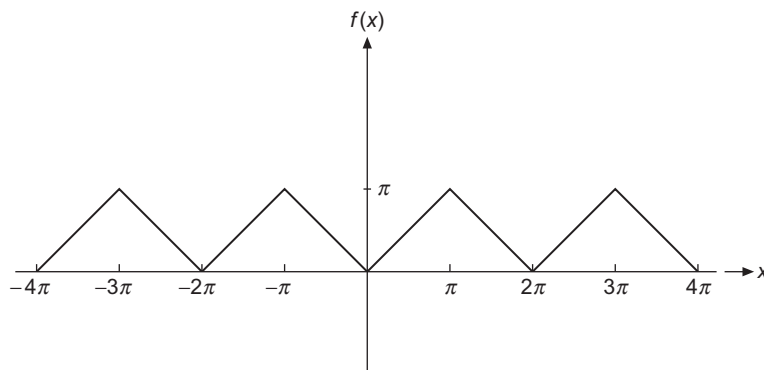


FIGURE 19.4 Triangular wave.

- As the inverse of integration, the operation of differentiation has placed an additional factor n in the numerator of each term. This reduces the rate of convergence and may, as in the first case mentioned, render the differentiated series divergent.
- In general, term-by-term differentiation is permissible if the series to be differentiated is uniformly convergent.

Summing Fourier Series

Often the most efficient way to identify the function represented by a Fourier series is simply to identify the expansion in a table. But if it is our desire to sum the series ourselves, a useful approach is to replace the trigonometric functions by their complex exponential forms, and then identifying the Fourier series as one or more power series in $e^{\pm ix}$.

Example 19.1.3 SUMMATION OF A FOURIER SERIES

Consider the series $\sum_{n=1}^{\infty} (1/n) \cos nx$, $x \in (0, 2\pi)$. Since this series is only conditionally convergent (and diverges at $x = 0$), we take

$$\sum_{n=1}^{\infty} \frac{\cos nx}{n} = \lim_{r \rightarrow 1} \sum_{n=1}^{\infty} \frac{r^n \cos nx}{n},$$

absolutely convergent for $|r| < 1$. Our procedure is to try forming power series by transforming the trigonometric functions into exponential form:

$$\sum_{n=1}^{\infty} \frac{r^n \cos nx}{n} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{r^n e^{inx}}{n} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{r^n e^{-inx}}{n}.$$

Now, these power series may be identified as Maclaurin expansions of $-\ln(1 - z)$, with $z = re^{ix}$ or re^{-ix} . From Eq. (1.97),

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{r^n \cos nx}{n} &= -\frac{1}{2} [\ln(1 - re^{ix}) + \ln(1 - re^{-ix})] \\ &= -\ln[(1 + r^2) - 2r \cos x]^{1/2}. \end{aligned}$$

Setting $r = 1$, we see that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{\cos nx}{n} &= -\ln(2 - 2 \cos x)^{1/2} \\ &= -\ln\left(2 \sin \frac{x}{2}\right), \quad (0 < x < 2\pi). \end{aligned} \tag{19.24}$$

Both sides of this expression diverge as $x \rightarrow 0$ and as $x \rightarrow 2\pi$.⁴ ■

⁴Note that the range of validity of Eq. (19.24) may be shifted to $[-\pi, \pi]$ (excluding $x = 0$) if we replace x by $|x|$ on the right-hand side.

Exercises

- 19.1.1** A function $f(x)$ (quadratically integrable) is to be represented by a **finite** Fourier series. A convenient measure of the accuracy of the series is given by the integrated square of the deviation,

$$\Delta_p = \int_0^{2\pi} \left[f(x) - \frac{a_0}{2} - \sum_{n=1}^p (a_n \cos nx + b_n \sin nx) \right]^2 dx.$$

Show that the requirement that Δ_p be minimized, that is,

$$\frac{\partial \Delta_p}{\partial a_n} = 0, \quad \frac{\partial \Delta_p}{\partial b_n} = 0,$$

for all n , leads to choosing a_n and b_n as given in Eqs. (19.2) and (19.3).

Note. Your coefficients a_n and b_n are independent of p . This independence is a consequence of orthogonality and would not hold if we expanded $f(x)$ in a power series.

- 19.1.2** In the analysis of a complex waveform (ocean tides, earthquakes, musical tones, etc.), it might be more convenient to have the Fourier series written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \alpha_n \cos(nx - \theta_n).$$

Show that this is equivalent to Eq. (19.1) with

$$\begin{aligned} a_n &= \alpha_n \cos \theta_n, & \alpha_n^2 &= a_n^2 + b_n^2, \\ b_n &= \alpha_n \sin \theta_n, & \tan \theta_n &= b_n/a_n. \end{aligned}$$

Note. The coefficients α_n^2 as a function of n define what is called the **power spectrum**. The importance of α_n^2 lies in their invariance under a shift in the phase θ_n .

- 19.1.3** A function $f(x)$ is expanded in an exponential Fourier series

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}.$$

If $f(x)$ is real, $f(x) = f^*(x)$, what restriction is imposed on the coefficients c_n ?

- 19.1.4** Assuming that $\int_{-\pi}^{\pi} [f(x)]^2 dx$ is finite, show that

$$\lim_{m \rightarrow \infty} a_m = 0, \quad \lim_{m \rightarrow \infty} b_m = 0.$$

Hint. Integrate $[f(x) - s_n(x)]^2$, where $s_n(x)$ is the n th partial sum, and use Bessel's inequality (Section 5.1). For our finite interval the assumption that $f(x)$ is square integrable ($\int_{-\pi}^{\pi} |f(x)|^2 dx$ is finite) implies that $\int_{-\pi}^{\pi} |f(x)| dx$ is also finite. The converse does not hold.

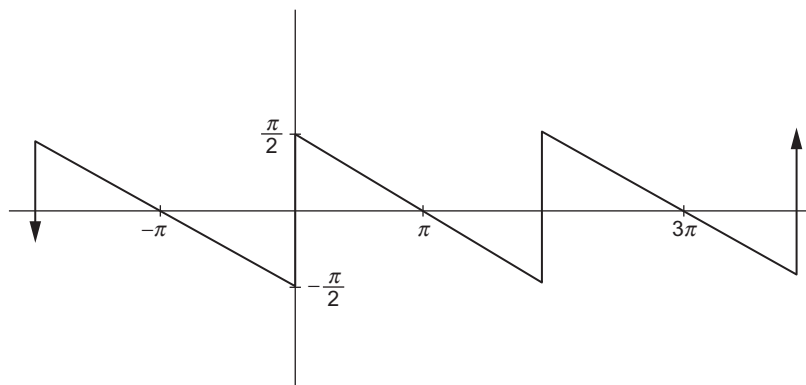


FIGURE 19.5 Reverse sawtooth wave.

19.1.5 Apply the summation technique of this section to show that

$$\sum_{n=1}^{\infty} \frac{\sin nx}{n} = \begin{cases} \frac{1}{2}(\pi - x), & 0 < x \leq \pi, \\ -\frac{1}{2}(\pi + x), & -\pi \leq x < 0. \end{cases}$$

This is the reverse sawtooth wave shown in Fig. 19.5.

19.1.6 Sum the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin nx}{n}$ and show that it equals $x/2$.

19.1.7 Sum the trigonometric series $\sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1}$ and show that it equals

$$\begin{cases} \pi/4, & 0 < x < \pi, \\ -\pi/4, & -\pi < x < 0. \end{cases}$$

19.1.8 Let $f(z) = \ln(1+z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} z^n}{n}$. This series converges to $\ln(1+z)$ for $|z| \leq 1$, except at the point $z = -1$.

(a) From the real parts show that

$$\ln \left(2 \cos \frac{\theta}{2} \right) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\cos n\theta}{n}, \quad -\pi < \theta < \pi.$$

(b) Using a change of variable, transform part (a) into

$$-\ln \left(2 \sin \frac{\theta}{2} \right) = \sum_{n=1}^{\infty} \frac{\cos n\theta}{n}, \quad 0 < \theta < 2\pi.$$

19.1.9 (a) Expand $f(x) = x$ in the interval $(0, 2L)$. Sketch the series you have found (right-hand side of ANS.) over $(-2L, 2L)$.

$$\text{ANS. } x = L - \frac{2L}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \left(\frac{n\pi x}{L} \right).$$

- (b) Expand $f(x) = x$ as a sine series in the **half** interval $(0, L)$. Sketch the series you have found (right-hand side of Ans.) over $(-2L, 2L)$.

$$\text{ANS. } x = \frac{4L}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} \sin\left(\frac{(2n+1)\pi x}{L}\right).$$

- 19.1.10** In some problems it is convenient to approximate $\sin \pi x$ over the interval $[0, 1]$ by a parabola $ax(1-x)$, where a is a constant. To get a feeling for the accuracy of this approximation, expand $4x(1-x)$ in a Fourier sine series $(-1 \leq x \leq 1)$:

$$f(x) = \begin{cases} 4x(1-x), & 0 \leq x \leq 1 \\ 4x(1+x), & -1 \leq x \leq 0 \end{cases} = \sum_{n=1}^{\infty} b_n \sin n\pi x.$$

$$\begin{aligned} \text{ANS. } b_n &= \frac{32}{\pi^3} \frac{1}{n^3}, & n \text{ odd,} \\ b_n &= 0, & n \text{ even.} \end{aligned}$$

This approximation is shown in Fig. 19.6.

- 19.1.11** Verify that $\delta(\varphi_1 - \varphi_2) = \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} e^{im(\varphi_1 - \varphi_2)}$ is a Dirac delta function by showing that it satisfies the definition,

$$\int_{-\pi}^{\pi} f(\varphi_1) \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} e^{im(\varphi_1 - \varphi_2)} d\varphi_1 = f(\varphi_2).$$

Hint. Represent $f(\varphi_1)$ by an exponential Fourier series.

- 19.1.12** Show that integration of the Fourier expansion of $f(x) = x$, $-\pi < x < \pi$, leads to

$$\frac{\pi^2}{12} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \cdots.$$

Note. The series for $f(x) = x$ was the subject of Example 19.1.1. Confirm that the change in the defined range from $[0, 2\pi]$ to $[-\pi, \pi]$ has no effect on the expansion.

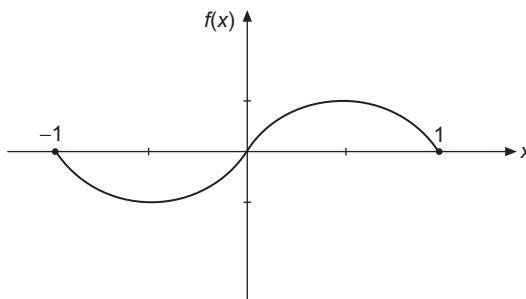


FIGURE 19.6 Parabolic approximation to sine wave.

- 19.1.13** (a) Assuming that the Fourier expansion of $f(x)$ is uniformly convergent, show that

$$\frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

This is Parseval's identity. Note that it is a completeness relation for the Fourier expansion.

(b) Given $x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n^2}$, $-\pi \leq x \leq \pi$,

apply Parseval's identity to obtain $\zeta(4)$ in closed form.

- (c) The condition of uniform convergence is not necessary. Show this by applying the Parseval identity to the square wave

$$\begin{aligned} f(x) &= \begin{cases} -1, & -\pi < x < 0 \\ 1, & 0 < x < \pi \end{cases} \\ &= \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2n-1)x}{2n-1}. \end{aligned}$$

- 19.1.14** Given

$$\varphi_1(x) \equiv \sum_{n=1}^{\infty} \frac{\sin nx}{n} = \begin{cases} -\frac{1}{2}(\pi + x), & -\pi \leq x < 0, \\ \frac{1}{2}(\pi - x), & 0 < x \leq \pi, \end{cases}$$

show by integrating that

$$\varphi_2(x) \equiv \sum_{n=1}^{\infty} \frac{\cos nx}{n^2} = \begin{cases} \frac{1}{4}(\pi + x)^2 - \frac{\pi^2}{12}, & -\pi \leq x \leq 0, \\ \frac{1}{4}(\pi - x)^2 - \frac{\pi^2}{12}, & 0 \leq x \leq \pi. \end{cases}$$

- 19.1.15** Given

$$\psi_{2s}(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n^{2s}}, \quad \psi_{2s+1}(x) = \sum_{n=1}^{\infty} \frac{\cos nx}{n^{2s+1}},$$

develop the following recurrence relations:

(a) $\psi_{2s}(x) = \int_0^x \psi_{2s-1}(x) dx,$

(b) $\psi_{2s+1}(x) = \zeta(2s+1) - \int_0^x \psi_{2s}(x) dx.$

Note. The functions $\psi_s(x)$ and $\varphi_s(x)$ of this and the preceding exercise are known as **Clausen functions**. In theory they may be used to improve the rate of convergence of a Fourier series. As is often the case, there is the question of how much analytical work we do and how much arithmetic work we demand that a computer do. As computers become steadily more powerful, the balance progressively shifts so that we are doing less and demanding that computers do more.

19.1.16 Show that $f(x) = \sum_{n=1}^{\infty} \frac{\cos nx}{n+1}$ may be written as

$$f(x) = \psi_1(x) - \varphi_2(x) + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2(n+1)},$$

where $\psi_1(x)$ and $\varphi_2(x)$ are the Clausen functions defined in [Exercises 19.1.14](#) and [19.1.15](#).

19.2 APPLICATIONS OF FOURIER SERIES

We present in this section two typical problems and a short table of useful Fourier series, followed by a substantial number of exercises that illustrate some of the techniques that arise in applications.

Example 19.2.1 SQUARE WAVE

One application of Fourier series, the analysis of a “square” wave ([Fig. 19.7](#)) in terms of its Fourier components, occurs in electronic circuits designed to handle sharply rising pulses. Suppose that our wave is defined by

$$\begin{aligned} f(x) &= 0, & -\pi < x < 0, \\ f(x) &= h, & 0 < x < \pi. \end{aligned} \tag{19.25}$$

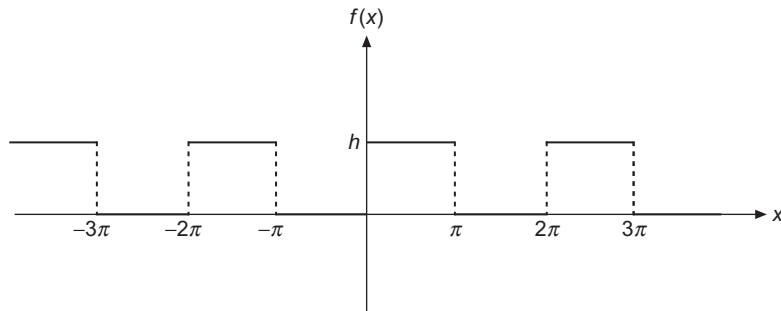


FIGURE 19.7 Square wave.

From Eqs. (19.2) and (19.3), we find

$$\begin{aligned}
 a_0 &= \frac{1}{\pi} \int_0^{\pi} h \, dt = h, \\
 a_n &= \frac{1}{\pi} \int_0^{\pi} h \cos nt \, dt = 0, \quad n = 1, 2, 3, \dots, \\
 b_n &= \frac{1}{\pi} \int_0^{\pi} h \sin nt \, dt = \frac{h}{n\pi} (1 - \cos n\pi) \\
 &= \begin{cases} \frac{2h}{n\pi}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}
 \end{aligned}$$

The resulting series is

$$f(x) = \frac{h}{2} + \frac{2h}{\pi} \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \dots \right). \quad (19.26)$$

Except for the first term, which represents an average of $f(x)$ over the interval $[-\pi, \pi]$, all the cosine terms have vanished. Since $f(x) - h/2$ is odd, we have a Fourier sine series. Although only the odd terms in the sine series occur, they fall only as n^{-1} . This **conditional convergence** is like that of the alternating harmonic series. Physically this means that our square wave contains a lot of **high-frequency components**. If the electronic apparatus will not pass these components, our square-wave input will emerge more or less rounded off, perhaps as an amorphous blob. ■

Example 19.2.2 FULL-WAVE RECTIFIER

As a second example, let us ask how well the output of a full-wave rectifier approaches pure direct current. Our rectifier may be thought of as passing the positive peaks of an incoming sine wave and inverting the negative peaks, as shown in Fig. 19.8. This yields

$$f(t) = \begin{cases} \sin \omega t, & 0 < \omega t < \pi, \\ -\sin \omega t, & -\pi < \omega t < 0. \end{cases} \quad (19.27)$$

Since $f(t)$ as defined here is even, no terms of the form $\sin n\omega t$ will appear. Again, from Eqs. (14.2) and (14.3), we have

$$\begin{aligned}
 a_0 &= -\frac{1}{\pi} \int_{-\pi}^0 \sin \omega t \, d(\omega t) + \frac{1}{\pi} \int_0^{\pi} \sin \omega t \, d(\omega t) \\
 &= \frac{2}{\pi} \int_0^{\pi} \sin \omega t \, d(\omega t) = \frac{4}{\pi},
 \end{aligned}$$

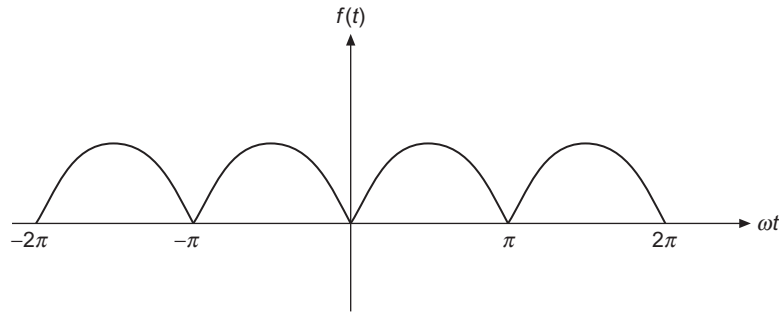


FIGURE 19.8 Full wave rectifier.

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \int_0^{\pi} \sin \omega t \cos n \omega t \, d(\omega t) \\
 &= \begin{cases} -\frac{2}{\pi} \frac{2}{n^2 - 1}, & n \text{ even,} \\ 0, & n \text{ odd.} \end{cases}
 \end{aligned}$$

Note that $[0, \pi]$ is not an orthogonality interval for both sines and cosines together and we do not get zero when n is even. The resulting series is

$$f(t) = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=2,4,6,\dots}^{\infty} \frac{\cos n \omega t}{n^2 - 1}. \quad (19.28)$$

The original frequency, ω , has been eliminated; in fact, all its odd harmonics are also absent. The lowest-frequency oscillation is 2ω . The high-frequency components fall off as n^{-2} , showing that the full-wave rectifier does a fairly good job of approximating direct current. Whether this good approximation is adequate depends on the particular application. If the remaining alternating current components are objectionable, they may be further suppressed by appropriate filter circuits. ■

These examples bring out two features characteristic of Fourier expansions:⁵

- If $f(x)$ has discontinuities, as in the square wave in [Example 19.2.1](#), we can expect the n th coefficient to be decreasing as $\mathcal{O}(1/n)$. Convergence is conditional only.
- If $f(x)$ is continuous (although possibly with discontinuous derivatives as in the full-wave rectifier of [Example 19.2.2](#)), we can expect the n th coefficient to be decreasing as $1/n^2$, that is, absolute convergence.

We close this section by providing, in [Table 19.1](#), a list of Fourier series that have been introduced either as examples or in the exercises of this chapter. More extensive lists can be found in the Additional Readings, particularly in the work by Oberhettinger, but also in the texts by Carslaw, Churchill, and Zygmund.

⁵G. Raisbeek, Order of magnitude of Fourier coefficients. *Am. Math. Mon.* **62**: 149 (1955).

Table 19.1 Some Fourier Series Used in This Text

Fourier Series	Reference
1. $\sum_{n=1}^{\infty} \frac{\sin nx}{n} = \begin{cases} -\frac{1}{2}(\pi + x), & -\pi \leq x < 0 \\ \frac{1}{2}(\pi - x), & 0 \leq x < \pi \end{cases}$	Exercise 19.1.5 Exercise 19.2.8
2. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin nx}{n} = \frac{x}{2}, \quad -\pi < x < \pi$	Exercise 19.1.6 Exercise 19.2.7
3. $\sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1} = \begin{cases} -\pi/4, & -\pi < x < 0 \\ +\pi/4, & 0 < x < \pi \end{cases}$	Exercise 19.1.7 Eq. (19.26)
4. $\sum_{n=1}^{\infty} \frac{\cos nx}{n} = -\ln \left[2 \sin \left(\frac{ x }{2} \right) \right], \quad -\pi < x < \pi$	Exercise 19.1.8(b) Eq. (19.24)
5. $\sum_{n=1}^{\infty} (-1)^n \frac{\cos nx}{n} = -\ln \left[2 \cos \left(\frac{x}{2} \right) \right], \quad -\pi < x < \pi$	Exercise 19.1.8(a)
6. $\sum_{n=0}^{\infty} \frac{\cos(2n+1)x}{2n+1} = \frac{1}{2} \ln \left[\cot \frac{ x }{2} \right], \quad -\pi < x < \pi$	Exercise 19.2.5

Exercises

- 19.2.1** Transform the Fourier expansion of a square wave, Eq. (19.26), into a power series. Show that the coefficients of x^1 form a **divergent** series. Repeat for the coefficients of x^3 .

Note. A power series cannot handle a discontinuity. These infinite coefficients are the result of attempting to beat this basic limitation on power series.

- 19.2.2** Derive the Fourier series expansion of the Dirac delta function $\delta(x)$ in the interval $-\pi < x < \pi$.

- What significance can be attached to the constant term?
- In what region is this representation valid?
- With the identity

$$\sum_{n=1}^N \cos nx = \frac{\sin(Nx/2)}{\sin(x/2)} \cos \left[\left(N + \frac{1}{2} \right) \frac{x}{2} \right],$$

show that your Fourier representation of $\delta(x)$ is consistent with Eq. (5.27).

- 19.2.3** Expand $\delta(x - t)$ in a Fourier series. Compare your result with the bilinear form of Eq. (5.27).

$$\begin{aligned}
 \text{ANS. } \delta(x-t) &= \frac{1}{2\pi} + \frac{1}{\pi} \sum_{n=1}^{\infty} (\cos nx \cos nt + \sin nx \sin nt) \\
 &= \frac{1}{2\pi} + \frac{1}{\pi} \sum_{n=1}^{\infty} \cos n(x-t).
 \end{aligned}$$

- 19.2.4** Show that integrating the Fourier expansion of the Dirac delta function ([Exercise 19.2.2](#)) leads to the Fourier representation of the square wave, [Eq. \(19.26\)](#), with $h = 1$.

Note. Integrating the constant term $(1/2\pi)$ leads to a term $x/2\pi$. What are you going to do with this?

- 19.2.5** Starting from the Fourier series given as lines 4 and 5 of [Table 19.1](#), show that:

$$\sum_{n=0}^{\infty} \frac{\cos(2n+1)x}{2n+1} = \frac{1}{2} \ln \left[\cot \frac{|x|}{2} \right].$$

- 19.2.6** Develop the Fourier series representation of

$$f(t) = \begin{cases} 0, & -\pi \leq \omega t \leq 0, \\ \sin \omega t, & 0 \leq \omega t \leq \pi. \end{cases}$$

This is the output of a simple half-wave rectifier. It is also an approximation of the solar thermal effect that produces “tides” in the atmosphere.

$$\text{ANS. } f(t) = \frac{1}{\pi} + \frac{1}{2} \sin \omega t - \frac{2}{\pi} \sum_{n=2,4,6,\dots}^{\infty} \frac{\cos n\omega t}{n^2 - 1}.$$

- 19.2.7** A sawtooth wave is given by

$$f(x) = x, \quad -\pi < x < \pi.$$

Show that

$$f(x) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx.$$

- 19.2.8** A different sawtooth wave is described by

$$f(x) = \begin{cases} -\frac{1}{2}(\pi + x), & -\pi \leq x < 0 \\ +\frac{1}{2}(\pi - x), & 0 < x \leq \pi. \end{cases}$$

Show that $f(x) = \sum_{n=1}^{\infty} (\sin nx/n)$.

- 19.2.9** A triangular wave ([Fig. 19.4](#)) is represented by

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ -x, & -\pi < x < 0. \end{cases}$$

Represent $f(x)$ by a Fourier series.

$$\text{ANS. } f(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots} \frac{\cos nx}{n^2}.$$

19.2.10 Expand

$$f(x) = \begin{cases} 1, & x^2 < x_0^2 \\ 0, & x^2 > x_0^2 \end{cases}$$

in the interval $[-\pi, \pi]$.

Note. This variable-width square wave is of some importance in electronic music.

19.2.11 A metal cylindrical tube of radius a is split lengthwise into two nontouching halves. The top half is maintained at a potential $+V$, the bottom half at a potential $-V$. See Fig. 19.9. Separate the variables in Laplace's equation and solve for the electrostatic potential for $r \leq a$. Observe the resemblance between your solution for $r = a$ and the Fourier series for a square wave.

19.2.12 A metal cylinder is placed in a (previously) uniform electric field, E_0 , with the axis of the cylinder perpendicular to that of the original field.

- (a) Find the perturbed electrostatic potential.
- (b) Find the induced surface charge on the cylinder as a function of angular position.

19.2.13 (a) Find the Fourier series representation of

$$f(x) = \begin{cases} 0, & -\pi < x \leq 0 \\ x, & 0 \leq x < \pi. \end{cases}$$

(b) From the Fourier expansion show that

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.$$

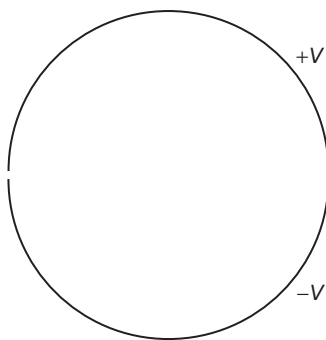


FIGURE 19.9 Cross section of split tube.

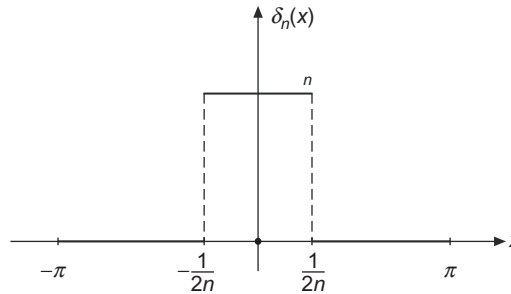


FIGURE 19.10 Rectangular pulse.

19.2.14 Integrate the Fourier expansion of the unit step function

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 1, & 0 < x < \pi. \end{cases}$$

Show that your integrated series agrees with [Exercise 19.2.13](#).

19.2.15 In the interval $(-\pi, \pi)$, $\delta_n(x) = \begin{cases} n, & |x| < 1/2n, \\ 0, & |x| > 1/2n. \end{cases}$

This wave form is the pulse shown in [Fig. 19.10](#).

- (a) Expand $\delta_n(x)$ as a Fourier cosine series.
- (b) Show that your Fourier series agrees with a Fourier expansion of $\delta(x)$ in the limit as $n \rightarrow \infty$.

19.2.16 Confirm the delta function nature of your Fourier series of [Exercise 19.2.15](#) by showing that for any $f(x)$ that is finite in the interval $[-\pi, \pi]$ and continuous at $x = 0$,

$$\int_{-\pi}^{\pi} f(x) [\text{Fourier expansion of } \delta_{\infty}(x)] dx = f(0).$$

19.2.17 (a) Show that the Dirac delta function $\delta(x - a)$, expanded in a Fourier sine series in the half-interval $(0, L)$ ($0 < a < L$) is given by

$$\delta(x - a) = \frac{2}{L} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi a}{L}\right) \sin\left(\frac{n\pi x}{L}\right).$$

Note that this series actually describes $-\delta(x + a) + \delta(x - a)$ in the interval $(-L, L)$.

- (b) By integrating both sides of the preceding equation from 0 to x , show that the cosine expansion of the square wave

$$f(x) = \begin{cases} 0, & 0 \leq x < a \\ 1, & a < x < L, \end{cases}$$

is

$$f(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi a}{L}\right) - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi a}{L}\right) \cos\left(\frac{n\pi x}{L}\right),$$

for $0 \leq x < L$.

- (c) Show that the term $\frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi a}{L}\right)$ is the average of $f(x)$ on $(0, L)$.

19.2.18 Verify the Fourier cosine expansion of the square wave, [Exercise 19.2.17\(b\)](#), by direct calculation of the Fourier coefficients.

19.2.19 (a) A string is clamped at both ends $x = 0$ and $x = L$. Assuming small-amplitude vibrations, we find that the amplitude $y(x, t)$ satisfies the wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}.$$

Here v is the wave velocity. The string is set in vibration by a sharp blow at $x = a$. Hence we have

$$y(x, 0) = 0, \quad \frac{\partial y(x, t)}{\partial t} = Lv_0 \delta(x - a) \text{ at } t = 0.$$

The constant L is included to compensate for the dimensions (inverse length) of $\delta(x - a)$. With $\delta(x - a)$ given by [Exercise 19.2.17\(a\)](#), solve the wave equation subject to these initial conditions.

$$\text{ANS. } y(x, t) = \frac{2v_0 L}{\pi v} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a}{L} \sin \frac{n\pi x}{L} \sin \frac{n\pi vt}{L}.$$

- (b) Show that the transverse velocity of the string $\partial y(x, t)/\partial t$ is given by

$$\frac{\partial y(x, t)}{\partial t} = 2v_0 \sum_{n=1}^{\infty} \sin \frac{n\pi a}{L} \sin \frac{n\pi x}{L} \cos \frac{n\pi vt}{L}.$$

19.2.20 A string, clamped at $x = 0$ and at $x = L$, is vibrating freely. Its motion is described by the wave equation

$$\frac{\partial^2 u(x, t)}{\partial t^2} = v^2 \frac{\partial^2 u(x, t)}{\partial x^2}.$$

Assume a Fourier expansion of the form

$$u(x, t) = \sum_{n=1}^{\infty} b_n(t) \sin \frac{n\pi x}{L}$$

and determine the coefficients $b_n(t)$. The initial conditions are

$$u(x, 0) = f(x) \quad \text{and} \quad \frac{\partial}{\partial t} u(x, 0) = g(x).$$

Note. This is only half the conventional Fourier orthogonality integral interval. However, as long as only the sines are included here, the Sturm-Liouville boundary conditions are still satisfied and the functions are orthogonal.

$$\text{ANS. } b_n(t) = A_n \cos \frac{n\pi vt}{L} + B_n \sin \frac{n\pi vt}{L},$$

$$A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx, \quad B_n = \frac{2}{n\pi v} \int_0^L g(x) \sin \frac{n\pi x}{L} dx.$$

- 19.2.21** (a) Let us continue the vibrating string problem in [Exercise 19.2.20](#). We assume now that the presence of a resisting medium will damp the vibrations according to the equation

$$\frac{\partial^2 u(x, t)}{\partial t^2} = v^2 \frac{\partial^2 u(x, t)}{\partial x^2} - k \frac{\partial u(x, t)}{\partial t}.$$

Introduce a Fourier expansion

$$u(x, t) = \sum_{n=1}^{\infty} b_n(t) \sin \frac{n\pi x}{L}$$

and again determine the coefficients $b_n(t)$. Take the initial and boundary conditions to be the same as in [Exercise 19.2.20](#). Assume the damping to be small.

- (b) Repeat, but assume the damping to be large.

ANS.

$$\begin{aligned} \text{(a) } b_n(t) &= e^{-kt/2} [A_n \cos \omega_n t + B_n \sin \omega_n t], \quad \omega_n^2 = \left(\frac{n\pi v}{L}\right)^2 - \left(\frac{k}{2}\right)^2 > 0, \\ A_n &= \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx, \quad B_n = \frac{2}{\omega_n L} \int_0^L g(x) \sin \frac{n\pi x}{L} dx + \frac{k}{2\omega_n} A_n. \\ \text{(b) } b_n(t) &= e^{-kt/2} [A_n \cosh \sigma_n t + B_n \sinh \sigma_n t], \quad \sigma_n^2 = \left(\frac{k}{2}\right)^2 - \left(\frac{n\pi v}{L}\right)^2 > 0, \\ A_n &= \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx, \quad B_n = \frac{2}{\sigma_n L} \int_0^L g(x) \sin \frac{n\pi x}{L} dx + \frac{k}{2\sigma_n} A_n. \end{aligned}$$

19.3 GIBBS PHENOMENON

The Gibbs phenomenon is an overshoot, a peculiarity of the Fourier series and other eigenfunction series at a simple discontinuity. An example is seen in [Fig. 19.2](#).

Partial Summation of Fourier Series

To better understand the Gibbs phenomenon we examine methods for the partial summation of Fourier series. This procedure is unlikely to lead to convenient solutions of practical

problems for which Fourier series are ideal, but it may provide insight that is needed for our present study.

We start from the Fourier series of a function $f(x)$ in exponential form, truncating it to retain terms only for $n \leq |r|$ and labeling the truncated expansion $f_r(x)$:

$$f_r(x) = \sum_{n=-r}^r c_n e^{inx}, \quad c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt.$$

Combining these equations in a way useful for the present discussion, we have

$$f_r(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \sum_{n=-r}^r e^{i(x-t)n} dt. \quad (19.29)$$

The summation in Eq. (19.29) is a geometric series. Using a result easily obtained from Eq. (1.96),

$$\sum_{n=-r}^r y^n = \frac{y^{-r} - y^{r+1}}{1 - y} = \frac{y^{r+\frac{1}{2}} - y^{-(r+\frac{1}{2})}}{y^{1/2} - y^{-1/2}},$$

we set $y = e^{i(x-t)}$, after which we can identify the resulting expression as a quotient of sine functions:⁶

$$\sum_{n=-r}^r e^{in(x-t)} = \frac{e^{i(r+\frac{1}{2})(x-t)} - e^{-i(r+\frac{1}{2})(x-t)}}{e^{i(x-t)/2} - e^{-i(x-t)/2}} = \frac{\sin[(r+\frac{1}{2})(x-t)]}{\sin \frac{1}{2}(x-t)}. \quad (19.30)$$

Inserting Eq. (19.30) into Eq. (19.29), we reach

$$f_r(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \frac{\sin[(r+\frac{1}{2})(x-t)]}{\sin \frac{1}{2}(x-t)} dt. \quad (19.31)$$

This is convergent at all points, including $t = x$. Equation (19.31) shows that the quantity

$$\frac{1}{2\pi} \frac{\sin[(r+\frac{1}{2})(x-t)]}{\sin \frac{1}{2}(x-t)}$$

is in the large- r limit a Dirac delta distribution.

Square Wave

For convenience of numerical calculation we consider the behavior of the Fourier series that represents the periodic square wave

$$f(x) = \begin{cases} \frac{h}{2}, & 0 < x < \pi, \\ -\frac{h}{2}, & -\pi < x < 0. \end{cases} \quad (19.32)$$

⁶This series also occurs in the analysis of a diffraction grating consisting of r slits.

This is essentially the square wave used in [Example 19.2.1](#), and we immediately see that its Fourier expansion is

$$f(x) = \frac{2h}{\pi} \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots \right). \quad (19.33)$$

Applying [Eq. \(19.31\)](#) to our square wave, we have

$$f_r(x) = \frac{h}{4\pi} \int_0^\pi \frac{\sin[(r + \frac{1}{2})(x - t)]}{\sin \frac{1}{2}(x - t)} dt - \frac{h}{4\pi} \int_{-\pi}^0 \frac{\sin[(r + \frac{1}{2})(x - t)]}{\sin \frac{1}{2}(x - t)} dt.$$

Making the substitution $x - t = s$ in the first integral and $x - t = -s$ in the second, we obtain

$$f_r(x) = \frac{h}{4\pi} \int_{-\pi+x}^x \frac{\sin(r + \frac{1}{2})s}{\sin \frac{1}{2}s} ds - \frac{h}{4\pi} \int_{-\pi-x}^{-x} \frac{\sin(r + \frac{1}{2})s}{\sin \frac{1}{2}s} ds. \quad (19.34)$$

It is important to note that both the integrals in [Eq. \(19.34\)](#) have the same integrand, and therefore have the same indefinite integral, which we denote $\Phi(t)$. We may therefore write

$$\begin{aligned} f(r) &= \frac{h}{4\pi} [\Phi(x) - \Phi(-\pi + x)] - \frac{h}{4\pi} [\Phi(-x) - \Phi(-\pi - x)] \\ &= \frac{h}{4\pi} [\Phi(x) - \Phi(-x)] - \frac{h}{4\pi} [\Phi(-\pi + x) - \Phi(-\pi - x)], \end{aligned} \quad (19.35)$$

where the second line of [Eq. \(19.35\)](#) is an obvious rearrangement of the first. However, this second line is useful because it shows that we can also write $f_r(x)$ as

$$f_r(x) = \frac{h}{4\pi} \int_{-x}^x \frac{\sin(r + \frac{1}{2})s}{\sin \frac{1}{2}s} ds - \frac{h}{4\pi} \int_{-\pi-x}^{-\pi+x} \frac{\sin(r + \frac{1}{2})s}{\sin \frac{1}{2}s} ds. \quad (19.36)$$

We are now ready to consider the partial sums in the vicinity of the discontinuity, $x = 0$. For small x , the denominator of the second integrand approaches -1 , and the second integral therefore becomes negligible in the limit $x \rightarrow 0$. On the other hand, the first integrand becomes large near $s = 0$, and the value of the first integral depends on the magnitudes of r and x . If we now introduce the new variables $p = r + \frac{1}{2}$ and $\xi = ps$, we have (noting that the integrand is an even function of s)

$$f_r(x) \approx \frac{h}{2\pi} \int_0^{px} \frac{\sin \xi}{\sin(\xi/2p)} \frac{d\xi}{p}. \quad (19.37)$$

Calculation of Overshoot

We are now prepared to make a computation of the Fourier series overshoot. From [Eq. \(19.37\)](#), we see that for any finite r , $f_r(0)$ will be zero, giving at $x = 0$ the average of the two square-wave values ($+h/2$ and $-h/2$). However (keeping r fixed), [Eq. \(19.37\)](#) also tells us that $f_r(x)$ will increase as px becomes nonzero, reaching a maximum when

$px = \pi$. This maximum, which we will shortly show constitutes an overshoot, will therefore occur at $x = \pi/p$, which is approximately $x = \pi/r$. We thus see that the location of the overshoot maximum will differ from $x = 0$ in a manner approximately inversely proportional to the number of terms taken in the Fourier expansion.

To estimate the maximum value of $f_r(x)$, we substitute $px = \pi$ into Eq. (19.37), which we then simplify by making the good approximation $\sin(\xi/2p) \approx \xi/2p$:

$$f_r(x_{\max}) = \frac{h}{2\pi} \int_0^{\pi} \frac{\sin \xi \, d\xi}{p \sin(\xi/2p)} \approx \frac{h}{\pi} \int_0^{\pi} \frac{\sin \xi}{\xi} d\xi. \quad (19.38)$$

If the upper limit of the final integral of Eq. (19.38) were replaced by infinity, we would have

$$\int_0^{\infty} \frac{\sin \xi}{\xi} d\xi = \frac{\pi}{2}, \quad (19.39)$$

a result found in Example 11.8.5. Note that this replacement would cause $f_r(x)$ to have the value $h/2$, which is the exact value of $f(x)$ for $x > 0$.

The integral we would have to add to that of Eq. (19.38) to obtain the infinite range is

$$\int_{\pi}^{\infty} \frac{\sin \xi}{\xi} d\xi = -\text{si}(\pi); \quad (19.40)$$

we have identified this integral as the sine integral function $\text{si}(x)$ introduced in Table 1.2 and plotted in Fig. 13.6. Thus,

$$\int_0^{\pi} \frac{\sin \xi}{\xi} d\xi = \frac{\pi}{2} + \text{si}(\pi). \quad (19.41)$$

The graph of $\text{si}(x)$ shows that $\text{si}(\pi) > 0$, indicating an overshoot. A direct demonstration that our integral is larger than $\pi/2$ can also be deduced by writing

$$\left(\int_0^{\infty} - \int_{\pi}^{3\pi} - \int_{3\pi}^{5\pi} - \dots \right) \frac{\sin \xi}{\xi} d\xi = \int_0^{\pi} \frac{\sin \xi}{\xi} d\xi. \quad (19.42)$$

The first integral on the left-hand side has value $\pi/2$, while each of those to be subtracted is negative (and therefore makes a further positive contribution).

A Gaussian quadrature or a power-series expansion and term-by-term integration yields

$$\frac{2}{\pi} \int_0^{\pi} \frac{\sin \xi}{\xi} d\xi = 1.1789797 \dots, \quad (19.43)$$

which means that the Fourier series tends to overshoot the positive corner of the square wave by some 18% and to undershoot the negative corner by the same amount. This behavior is illustrated in Fig. 19.11. The inclusion of more terms (increasing r) does nothing to

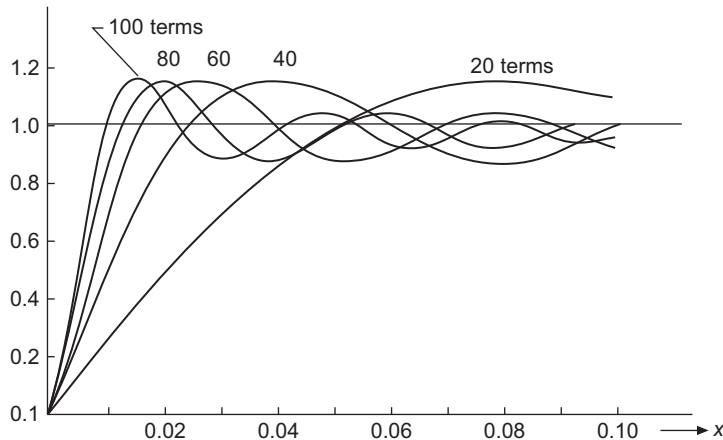


FIGURE 19.11 Square wave: Gibbs phenomenon.

remove this overshoot but merely moves it closer to the point of discontinuity. The overshoot is the Gibbs phenomenon, and because of it the Fourier series representation may be highly unreliable for precise numerical work, especially in the vicinity of a discontinuity.

The Gibbs phenomenon is not limited to the Fourier series. It occurs with other eigenfunction expansions. For more details, see W. J. Thompson, Fourier series and the Gibbs phenomenon, *Am. J. Phys.* **60**: 425 (1992).

Exercises

- 19.3.1** With the partial-sum summation techniques of this section, show that at a discontinuity in $f(x)$ the Fourier series for $f(x)$ takes on the arithmetic mean of the right- and left-hand limits:

$$f(x_0) = \frac{1}{2}[f(x_0 + 0) + f(x_0 - 0)].$$

In evaluating $\lim_{r \rightarrow \infty} s_r(x_0)$, you may find it convenient to identify part of the integrand as a Dirac delta function.

- 19.3.2** Determine the partial sum, s_n , of the series in Eq. (19.33) by using

$$(a) \quad \frac{\sin mx}{m} = \int_0^x \cos my \, dy, \quad (b) \quad \sum_{p=1}^n \cos(2p-1)y = \frac{\sin 2ny}{2 \sin y}.$$

Do you agree with the result given in Eq. (19.40)?

- 19.3.3** (a) Calculate the value of the Gibbs phenomenon integral

$$I = \frac{2}{\pi} \int_0^{\pi} \frac{\sin t}{t} dt$$

by numerical quadrature accurate to 12 significant figures.

- (b) Check your result by (1) expanding the integrand as a series, (2) integrating term by term, and (3) evaluating the integrated series. This calls for double-precision calculation.

ANS. $I = 1.178979744472$.

Additional Readings

Carslaw, H. S., *Introduction to the Theory of Fourier's Series and Integrals*, 2nd ed. London: Macmillan (1921); 3rd ed., paperback, Dover (1952). This is a detailed and classic work; includes considerable discussion of Gibbs phenomenon in chapter IX.

Churchill, R. V., *Fourier Series and Boundary Value Problems*, 5th ed., New York: McGraw-Hill (1993). Discusses uniform convergence in Section 38.

Jeffreys, H., and B. S. Jeffreys, *Methods of Mathematical Physics*, 3rd ed. Cambridge: Cambridge University Press (1972). Termwise integration of Fourier series is treated in section 14.06.

Kufner, A., and J. Kadlec, *Fourier Series*. London: Iliffe (1971). This book is a clear account of Fourier series in the context of Hilbert space.

Lanczos, C., *Applied Analysis*. Englewood Cliffs, NJ: Prentice-Hall (1956), reprinted, Dover (1988). The book gives a well-written presentation of the Lanczos convergence technique (which suppresses the Gibbs phenomenon oscillations). This and several other topics are presented from the point of view of a mathematician who wants useful numerical results and not just abstract existence theorems.

Oberhettinger, F., *Fourier Expansions; A Collection of Formulas*. New York: Academic Press (1973).

Zygmund, A., *Trigonometric Series*. Cambridge: Cambridge University Press (1988). The volume contains an extremely complete exposition, including relatively recent results in the realm of pure mathematics.